Math 257 Definitions and Theorems

Week 9: Modules 30-32

Fall 2023

30 Properties of Eigenvectors and Eigenvalues

Definition 30.1. Let A be an $n \times n$ matrix and let λ be an eigenvalue of A.

- The **algebraic multiplicity** of λ is its multiplicity as a root of the characteristic polynomial, that is, the largest integer k such that $(t-\lambda)^k$ divides $p_A(t)$.
- The **geometric multiplicity** of λ is the dimension of the eigenspace $\text{Eig}_\lambda(A)$ of λ.

Theorem 30.1. *Let A be an $n \times n$ matrix and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be eigenvectors of A corresponding to different eigenvalues. Then $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are linearly independent.*

Definition 30.2. Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}$$

The **trace** of A is the sum of the diagonal entries of A; that is

$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Theorem 30.2. *Let A be $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then*

- $tr(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.
- $\det(A) = \lambda_1 \cdot \lambda_2 \cdot \cdots \cdot \lambda_n$.

Theorem 30.3. *Let A be a 2×2-matrix. Then the characteristic polynomial of A is*

$$p(\lambda) = \lambda^2 - tr(A)\lambda + \det(A).$$

31 Markov matrices

Definition 31.1. An $n \times n$ matrix A is a **Markov matrix** (or: stochastic matrix) if it has only non-negative entries, and the entries in each column add up to 1.
A vector in $\mathbb{R}^n$ is a **probability vector** (or: stochastic vector) if has only non-negative entries, and the entries add up to 1.

Theorem 31.1. *Let A be a Markov matrix. Then*

- *1 is an eigenvalue of A and every other eigenvalue λ of A satisfies $|\lambda| \leq 1$.*
- *If A has only positive entries, then any other eigenvalue satisfies $|\lambda| < 1$.*

Definition 31.2. A **stationary** probability vector of a Markov matrix is probability vector $\mathbf{v}$ that is an eigenvector of A corresponding to the eigenvaule 1.

Theorem 31.2. *Let A be an $n \times n$-Markov matrix with only positive entries and let $\mathbf{z} \in \mathbb{R}^n$ be a probability vector. Then*

$$\mathbf{z}_\infty := \lim_{k \to \infty} A^k \mathbf{z} \text{ exists,}$$

and $\mathbf{z}_\infty$ is a stationary probability vector of A (ie. $A\boldsymbol{z}_\infty = \boldsymbol{z}_\infty$).

32 Diagonalization

Definition 32.1. A square matrix A is said to be **diagonalizable** if there is a invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Theorem 32.1. *Let A be an $n \times n$ matrix that has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ with associated eigenvalues $\lambda_1, \ldots, \lambda_n$. Then A is diagonalizable as PDP^{-1}, where*

$$P = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v_n} \end{bmatrix} \text{ and } D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Definition 32.2. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form an **eigenbasis** of $n \times n$ matrix A if $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form a basis of $\mathbb{R}^n$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are all eigenvectors of A.

Theorem 32.2. *The following are equivalent for $n \times n$ matrix A:*

- *A has an eigenbasis.*
- *A is diagonalizable.*
- *The geometric multiplicities of all eigenvalues of A sums up to n.*

Theorem 32.3. *Let A be $n \times n$ matrix and let $\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an eigenbasis of A. Then there is a diagonal matrix D such that*

$$A = I_{\mathcal{E}_n \mathcal{B}} D I_{\mathcal{B}\mathcal{E}_n}.$$